Video-Informed Scenario Reconstruction for Continuous Robotic Manipulation

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Abstract

Eight personally recorded phone videos capture a marked building block from synchronized overhead and oblique views. The 16 curated demonstrations deliberately include ordinary feeds, pauses and resumption, orientation corrections and already-completed tasks. This collection supplies starting configurations, correction states and completed-state controls for teaching a robot when to move, reorient or preserve an object. We reconstruct these observations as Franka Panda scenarios and generate robot-action supervision with a geometric expert. A separate visual adapter follows features extracted from the human footage. Adaptation of released X-VLA weights achieves 17/40 strict completions near recorded configurations. Diversifying destinations raises success on held-out combinations of familiar starts and goals from 0/12 to 10/12. Loading-specific teaching then enables a second placement task. Two frozen policies execute feeding followed by loading in a continuous simulated world. Across 36 prespecified paired draws, qualified workflow success is 25/36 natively (69.4%, exact 95% confidence interval 51.9–83.7%) and 31/36 with controller assistance (86.1%, 70.5–95.3%). The experiments connect task coverage, visual cues and grasp acquisition to distinct failures, providing a practical route from personally collected observations to measurable robot behaviour.

1 Introduction

Industrial deployment makes transfer between workstations an economic and engineering objective. Humanoid's reported proof of concept with Schaeffler involved picking metallic bearing rings from cluttered bins, transferring them to a buffer table and moving between stations. The account identifies reuse across ring types and autonomous regrasp as intended advantages, with teleoperated data from a physical twin used to adapt a pretrained VLA [1]. This motivates a practical question: can demonstrations teach reusable task structure that reduces additional teaching when the destination, fixture or initial object state changes?

We choose *feeding* and *loading* as a controlled abstraction of this parts-handling workflow. Feeding moves an orange DUPLO block bearing a blue numeral 2 into a tabletop staging square. Loading transfers it into a shallow fixture. Both tasks require acquisition, transport, orientation and release, providing a controlled test of skill reuse across destinations. The blue numeral distinguishes otherwise similar orientations. Personal phone recordings supply observed configurations and correction examples.

These experiments measure bounded layout and goal changes, including preservation of completed work and continuous task execution.

We report three findings. Broader source teaching improves local manipulation, and varied goals expand destination coverage. Loading requires task-specific supervision, with task selection and grasp acquisition responding to different interventions. Finally, frozen feeding and loading policies execute in one continuous simulated world with externally supplied switching. Direct visual retargeting and online candidate ranking provide additional comparisons.

2 Related work

X-VLA uses a flow-matching action model with embodiment-specific soft prompts and projections to accommodate heterogeneous robot data [2]. Its released weights and configurable robot-action interface make it a practical base for testing adaptation with limited task data. Human video can contribute at different levels: R3M learns reusable visual representations, whereas MimicPlay learns high-level plans from human play and uses robot demonstrations for low-level control [3, 4]. These distinctions motivate separating observations, task intentions and executable motor supervision.

Corrective imitation can address the distribution shift induced by a policy's own actions [5]. Here, expert continuations are admitted only when they execute successfully from the relevant state. The admitted continuations supply supervised action targets. Simulation uses robosuite and MuJoCo [6]. Simulator truth supports teacher construction and evaluation, with policy inputs specified below.

3 Data and system

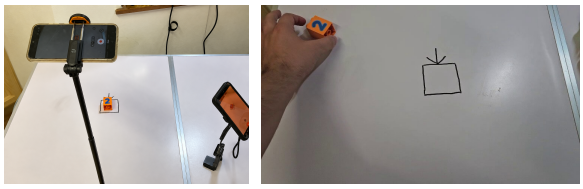
3.1 Recordings and coordinate conventions

Eight original phone recordings form four synchronized camera pairs: an approximately overhead view and an oblique view. The curated collection contains seven normal feeds, four awkward or pause–resume feeds, three orientation corrections

and two already-completed examples. The source archive also retains one reviewed attempt excluded from the curated set. Fifteen of the 16 curated examples contain sufficient observations for scenario reconstruction.

Claps, common object/hand events and stationary intervals support take boundaries and synchronization. Repositioning between takes is excluded from demonstrations. Classical colour and geometry processing extracts image-space block position, target boundaries and directed numeral orientation, with visibility/confidence masks. The upright primary view defines orientation. Missing or occluded measurements remain unknown. Source timestamps and recording identities connect each derived scenario to its footage.

Figure 1 shows the two-phone setup and an original demonstration frame. Target-relative coordinates map to a 100 mm virtual square. The simulation uses provisional dimensions of $50 \times 35 \times 30$ mm and mass of 80 g. Physical scale, camera depth and mass remain uncalibrated.



(a) Recording setup (b) Original demonstration

Figure 1: Two complementary views capture the marked block and target. (a) Setup photograph with AI-assisted background cleanup. (b) Original overhead video frame showing the demonstrator gripping the block.

3.2 Supervision, policy families and controller

The geometric teacher uses simulator object/goal states to generate approach, grasp, lift, transport, orientation, placement and release motions. It supplies robot RGB/action pairs in video-informed scenes. In an initial feasibility comparison, it completes 13/15 reconstructed personal scenarios and 8/15 independently sampled synthetic scenarios. These two panels use different start distributions.

We name the model families by the teaching change they test (Table 1). **Stream** uses dense causal robot video. **Chunks** adds single-phase and combined-phase examples plus a green held-object cue to make action structure explicit. **Anchors** compresses temporal supervision into windows of up to four seconds, preserving endpoints and gripper transitions. **Repair** adds successful expert continuations from states visited by Anchors. These variants test whether temporal structure and targeted correction improve execution. The held cue in the latter three families includes privileged simulator annotations.

The dense phase catalogue contains 316 semantic chunks. Temporal abstraction expands these into 388 windows, each with two camera orders. Sparse observation history samples approximately once per second, while future targets retain at most 30 action anchors and their physical timing. History features are mean-pooled with time/camera information. These chunks reuse 18 robot trajectories derived from ten personal and eight synthetic sources.

Each temporal family has prompt-only, low-rank and full-joint adaptation branches. Prompt-only training updates the active embodiment's soft-prompt entries. Low-rank training updates rank-eight LoRA adapters, the active soft prompts, action encoder/decoder entries and time/held-state projection. Full-joint training also updates the shared vision-language backbone and action transformer. Stream, Chunks and Anchors use 1,000 prompt updates, a separate 1,000-update prompt continuation, and 3,000 updates each for low-rank and full-joint branches.

A separate **Source** family tests broader source coverage through the native observation interface. It starts from released X-VLA-Pt weights and trains on all six designated source takes. Its 439 action windows receive 48,000 presentations, balanced at 8,000 per source. **Fixed**, **Range** and **Recovery** then compare original goals, varied goals and additional learner-state corrections from the same Source endpoint. **Switch** introduces loading and counterbalances fixture visibility across tasks. **Retain** adds acquisition and completed-state preservation examples.

These native-interface policies receive two current RGB views, an RGB-derived object crop, proprioception and a task instruction. They predict absolute end-effector position, 6D rotation and a gripper command. Each chunk contains 30 actions at 20 Hz, giving 1.5 s before reobservation. An operational-space controller converts these requests into robot motion.

Retain uses AdamW with $\beta = (0.9, 0.95)$, effective batch four, weight decay 0.01 on shared weights, and bfloat16 autocast with float32 weights. Learning rates are 10^{-5} for vision-language weights/prompts and 10^{-4} for the action core/projections. Its 12,000 sample presentations balance feeding and loading, including prior-skill rehearsal and acquisition/preservation examples.

3.3 Task success and experimental units

We distinguish three measurements. *Supported pickup* requires bilateral contact and at least 10 mm lift for three consecutive control samples. The final workflow audit additionally requires tilt within 10°. *Operational placement* requires containment, appropriate support height, an upright marked face, release and stability. *Strict completion* also requires centre error $\leq$ 15 mm and directed yaw error $\leq 10°$. Height tolerance is 2 mm, and loading must be seated. A good yaw score alone is insufficient when the block is tipped.

The primary endpoint of the continuous evaluation is *qualified workflow success*: both stages pass strict completion, no forbidden gripper/table-or-fixture contact is recorded, joint margins remain nonnegative, and any assistance primitive finishes successfully. Contact qualification uses sampled logs.

Exploratory panels report finite counts over reused source families and inspected scenes. Goal-diversity intervals use a source-family bootstrap over six families. The final reliability panel uses 36 prespecified scenario–seed draws,

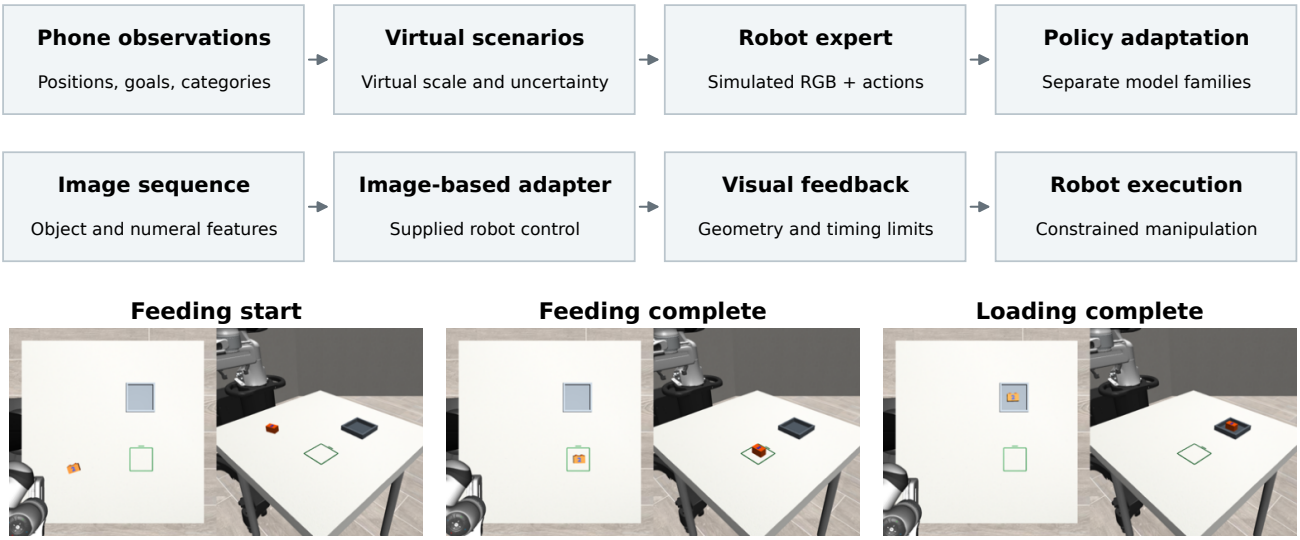


Figure 2: Two implemented routes from personal observations to manipulation. In the learning route, video specifies scenarios and the simulator expert supplies robot-action labels. The visual adapter instead follows image descriptors with supplied robot control and timing assumptions. Below: three original simulator observations from a successful development workflow, showing the feeding start, completed feeding and completed loading. The task switches externally within one continuous simulated world.

Table 1: Model families, training inputs and updated parameters. The family names identify data choices. Adaptation branches identify which weights change. Counts denote optimizer updates.

Family and purpose	Training data and initialization	Updated parameters and budget
Stream / Chunks / Anchors: temporal structure	Generated robot RGB/action sequences in reconstructed scenes. Dense history, semantic chunks or sparse history with timed action anchors.	Separate prompt-only, rank-eight LoRA and full-joint branches as defined in the text. Prompt branches receive 1,000 updates each, LoRA and full-joint branches 3,000 each.
Repair: learner-state recovery	Continues the corresponding Anchors branch with original data and successful same-state expert continuations.	Retains its parent's trainable modules. The full-joint endpoint evaluated here receives 500 additional updates.
Source: broader source coverage	Fresh released X-VLA-Pt, six personal source takes represented by generated robot trajectories.	Active prompts/action projections for 1,000 updates, then joint training of shared vision-language/action weights with a 1,000-update learning-rate ramp. Checkpoints at 3,000 and 12,000 updates.
Fixed / Range / Recovery: goal and correction coverage	Same Source-12k initializer. Six original-goal trajectories, 12 varied-goal trajectories, or six nominal plus four qualified recovery trajectories.	Full shared vision-language/action weights and active prompt/action projections, jointly for 3,000 updates. Batch four, matched source exposure, one fitting seed per condition.
Switch / Retain: task selection and preservation	Mixed feeding/loading data, counterbalanced fixture visibility, then acquisition/preservation examples with feeding rehearsal.	Full shared vision-language/action weights and active prompt/action projections. Separate 3,000-update continuations. Retain is the final loading checkpoint.
Continuous workflow	Frozen Range feeding and Retain loading checkpoints.	Weights remain fixed throughout evaluation. The assisted condition changes controller execution.

with exact 95% binomial intervals for per-mode success [7]. Paired endpoint-error intervals use 10,000 bootstrap resamples. These intervals describe scenario variation with fixed fitted policies.

4 Experiments and findings

4.1 What broader teaching changed

Two early evaluations exposed the limits of the initial adaptations. A 400-trial synthetic feeding/loading panel crossed five model conditions with guarded and unguarded execution. No strict success was recorded. Guarded runs were constrained

Table 2: Native-interface comparison. Entries are counts over six source families. Preservation measures whether an already-completed state remains correct.

Measure	Released	Source
Near-source pickup	0/40	31/40
Near-source operational placement	0/40	22/40
Near-source strict completion	0/40	17/40
Broader-start strict completion	0/20	0/20
Completed-state preservation	1/8	8/8

by a 30 s wall-clock cap, making them unsuitable for a clean estimate of the guard's effect. A separate 775-rollout panel evaluated five models on 155 matched feeding scenes, including reconstructed, nearby and independently synthetic starts. All strict passes began already complete. Repair showed partial grasps and placements while retaining orientation failures.

Source improves substantially near the recorded configurations (Table 2). It grasps in 31/40 cases, achieves operational placement in 22/40 and strict completion in 17/40, versus zero for the released native-interface baseline. Both fail all 20 broader-start cases. Already-completed controls reveal a separate ability: the adapted policy preserves 8/8 strict endpoints versus 1/8 for the released model.

A matched continuation comparison isolates useful teaching variation more directly. Fixed and Range share an initializer, optimizer budget and source exposure, with six and 12 unique teacher trajectories respectively. Range obtains 10/12 strict successes on withheld combinations of familiar starts and destinations, compared with 0/12 for Fixed (Fig. 3). Both feeding-trained variants score 0/12 on fixture loading. The full sweep contains separate panels of 136 baseline, 144 goal/task and 16 recovery episodes.

Adding four verified recovery trajectories from two source families also fails to solve a quarantined four-state recovery panel: all four compared endpoints score 0/4. Further direct recovery attempts show why teacher qualification matters. An expert continuation can place inside the fixture with 1.22 mm centre error and 0.016° yaw error yet still fail because the block lies on its side. Teacher admission therefore checks upright support as well as centre and yaw error.

4.2 Task selection and grasp acquisition are different failures

Loading-specific supervision enables a new destination. With the same additional 3,000-update budget, a feeding-only continuation remains at 0/4 source-loading and 0/2 withheld task-combination successes. Mixed feeding/loading teaching reaches 3/4 and 2/2, respectively, while strict feeding retention falls from 5/6 to 4/6. Stable in-square feeding placement remains 5/6. The additional failure is a centre-error threshold miss.

Initial task-switching tests show that fixture appearance can dominate the instruction. Counterbalancing visible/hidden fixture images across both tasks raises requested position/footprint attainment from 8/16 to 16/16 and strict completion from 8/16 to 12/16. Strict feeding rises from 4/8 to 8/8. Strict loading remains 4/8. In the four remaining loading failures, the block tips during approach, before grasp closure. It reaches the correct destination approximately 90° from upright. The failure begins during acquisition.

Acquisition/preservation teaching changes both pickup and completed-state behaviour. Retain preserves 8/8 completed-state controls and achieves 4/8 on the local loading-initiation panel. Its acquisition-focused parent achieves 5/8 and preserves 0/8. In two diagnosed failures, support is lost 1.6 s before opening while the gripper is still commanded closed. Lowering the saved wrist trajectory by 5 mm yields upright lifts of approximately 142 and 140 mm in bounded replay, motivating a controller-level acquisition correction. The replay endpoint is upright pickup.

4.3 Visibility and language diagnostics

Image resizing and occlusion can remove orientation detail. In one release-state audit, a 512×512 view becomes 224×224, leaving only a tiny visible fragment of the blue numeral. The object crop is absent at that instant. A 108-prediction intervention enlarges the cue and changes encoder features. The approximately 80° wrist-target error and opening waypoint persist. Action-level use of the numeral remains unresolved.

Simple instructions also fail as a general motor interface. Across 180 fixed-input predictions from the released model and two historical adaptations, each model produces zero of 18 required paired axis reversals under opposite-direction commands. Counterbalanced teaching subsequently improves feeding/loading destination selection.

Replanning and phase supervision. On Repair, reducing the execution horizon from 30 to ten actions before replanning changes recorded grasps from 2/12 to 0/12, with zero strict successes in either condition. An offline oracle over 48 proposals improves some approaches. Grasp-feasible states remain 1/6. An observable-phase supervisor blocks legitimate grasp attempts and fails qualification. A causal phase predictor also fails its early-release check. Completion observers fail their recall gates and receive no authority to advance task phases. These negative diagnostics motivate preserving separate evidence for proposal quality, observer reliability and control constraints.

4.4 Direct visual retargeting

A separate investigation extracts object paths, hand/pinch observations and frozen visual features directly from the original phone frames. On 526 sampled images, 440 centre detections pass the confidence threshold and hands are detected in 175 frames. Pooled cross-view take retrieval succeeds in only 1/16 cases. Denser sampling recovers additional visible motion. Metric lift height remains underdetermined.

Calibration sensitivity confirms the ambiguity: all 297 tested intrinsic settings per camera admit a target-corner fit

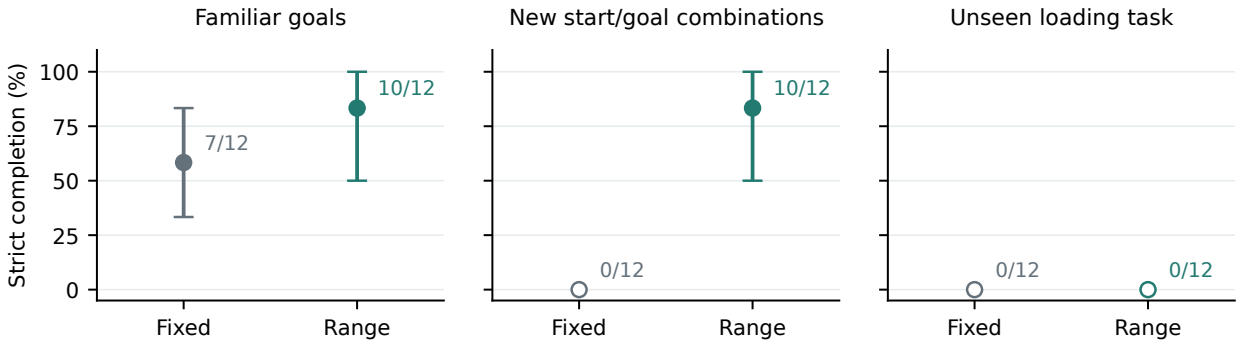


Figure 3: Range improves new combinations of familiar starts and destinations. Both feeding-trained variants score 0/12 on fixture loading. Points summarize 12 trials across six source families. Bars are exploratory 95% source-family bootstrap intervals from 10,000 resamples with fixed fitted models. All-zero cells have degenerate bootstrap intervals, omitted from the plot. Updates and family exposure are matched across six Fixed and 12 Range training trajectories.

within the assumed 5 px allowance, while implying substantially different off-plane geometry. The robot adapter uses image descriptors, a supplied virtual scale, calibrated simulator cameras and constrained visual servoing.

On one inspected 7.6 s correction take, five initial sequence/control executions grasp, lift and release. All miss the task goal by 33.7–35.5 mm. After registering the video endpoint to the fixed task goal, all five conditions—ordered sequence, endpoint-only, temporal shuffle, static reference and exact repeat—pass strict completion. The shared acquisition prefix, fourfold temporal slowdown and supplied release timing remain part of the adapter. A subsequent 736-observation audit finds depth-assumption bias in trajectory-fidelity scores.

4.5 Continuous feeding followed by loading

The final workflow uses frozen Range for feeding and Retain for loading. A single reset initializes the scene. Each stage receives 1,436 policy actions and 24 settling/scoring ticks at 20 Hz. Switching is externally timed and occurs regardless of success. The live robot, object and controller states carry across the boundary. An initial four-case development panel succeeds in 4/4 workflows per mode, motivating the larger prespecified evaluation.

The assisted condition changes only loading execution. It applies a frozen 60-action acquisition-height correction while retaining native gripper commands, followed when applicable by a five-action open-gripper clearance primitive. This deterministic assistance is separate from the candidate-ranking service described below. Earlier development tests had already shown a trade-off: an acquisition correction raised loading success from 4/8 to 8/8 and introduced forbidden contacts in 4/8 assisted cases. A broader 40-episode qualification also failed its complete gate. The later evaluation retains contact and joint-limit qualification alongside endpoint completion.

The reliability panel samples 36 draws from an equal-probability mixture of two recorded scenes, with independent uniform perturbations of ±5 mm in each tabletop direction and ±10° yaw. Both scenes originate from one camera pair and had been inspected previously. The realized allocation is 22/14. Each draw is executed once in each mode, with counterbalanced order, shared scenario/seed settings and no outcome-based retries. All 72 workflows and 144 stages complete.

Table 3: Continuous workflow outcomes. Each condition has 36 draws. Qualified success is the intersection of strict task completion and engineering qualification.

Outcome	Native	Assisted
Qualified success	25/36	31/36
95% exact interval (%)	51.9–83.7	70.5–95.3
Both strict stages completed	30/36	31/36
Engineering-qualified	30/36	35/36
Any forbidden contact recorded	6/36	1/36
Negative joint margin	0/36	1/36

Qualified success is 25/36 natively and 31/36 with assistance (Table 3). The observed difference is 16.7 percentage points: 24 pairs succeed in both modes, seven only with assistance, one only natively and four in neither. Five assisted-only wins preserve terminal completion while removing native contact violations. Two gain terminal completion. Assistance also loses one native terminal success.

Loading errors are heavy-tailed. Median position error is 6.44 mm natively and 4.80 mm assisted, whereas means are 37.88 and 30.21 mm. Median yaw error is 1.05° and 0.48°. Means are 5.89° and 5.19°. Maximum errors reach 354.62 mm and 125.23°. Mean paired changes are −7.667 mm and −0.706°, with uncertainty shown in Fig. 4. The exploratory position-change interval spans zero, while the yaw-change interval is entirely negative. Small typical errors coexist with occasional large failures.

The safety trade-off remains material. One assisted workflow has contacts and a minimum logged joint margin of −0.03106 rad. Across both modes there are 185 sampled forbidden-contact records in seven stages. One pair also diverges in upstream feeding images before loading, despite matched random-noise packets. Its loading handover differs, so 35/36 pairs provide exact upstream-controlled loading comparisons. Both outcomes for that draw succeed and remain in every reported denominator.

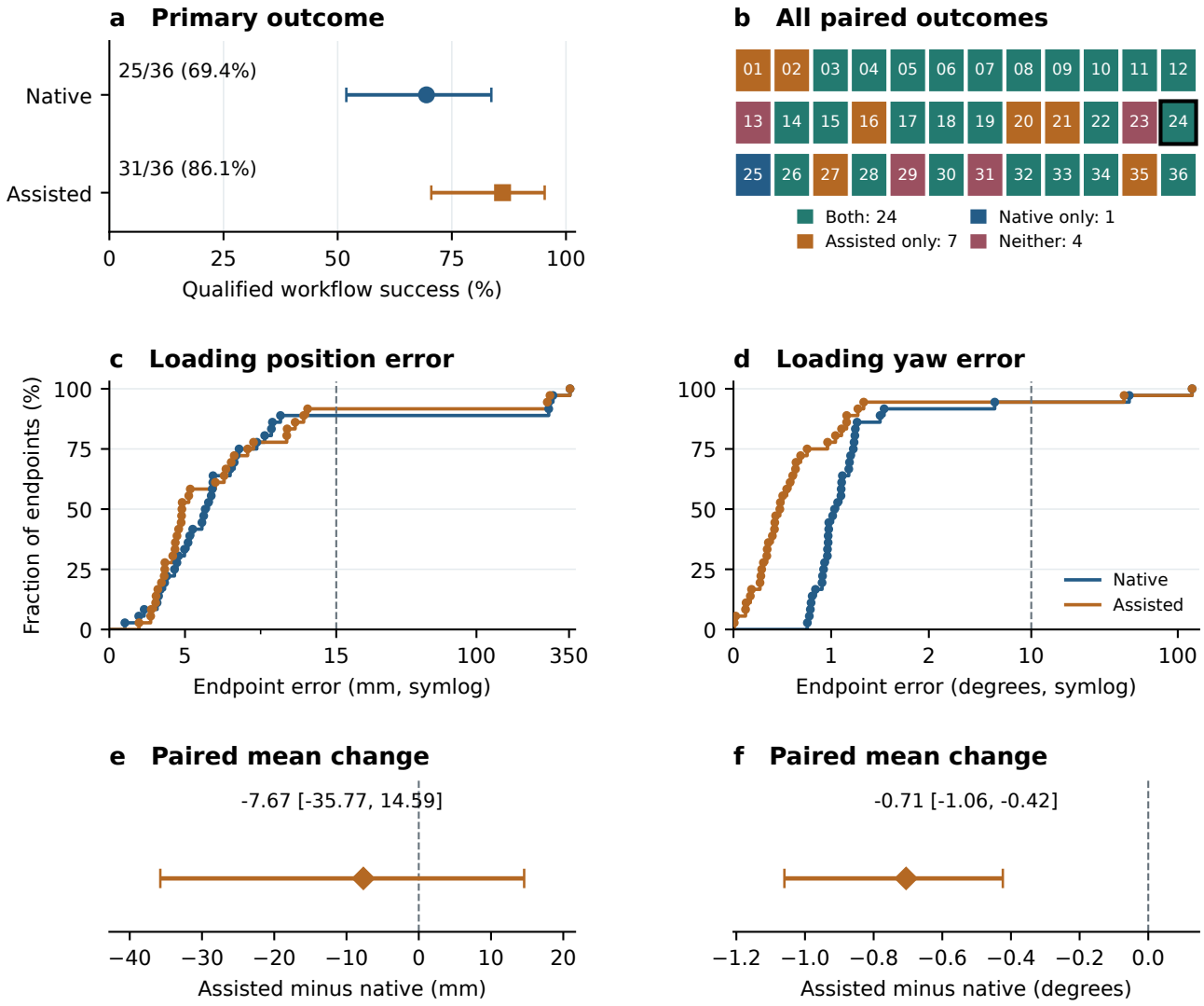


Figure 4: Continuous-workflow evaluation, retaining all 36 pairs and all failures. (a) Qualified success with marginal exact 95% confidence intervals. (b) Paired outcomes. The outlined draw has an upstream input discrepancy. (c,d) Complete loading endpoint-error distributions, with strict scalar tolerances shown as dashed lines. (e,f) Mean assisted-minus-native loading error with exploratory paired-bootstrap 95% intervals. Negative values favour assistance. Intervals describe the sampled simulator distribution with fixed policies.

4.6 Candidate ranking and online execution

A separate supervisory interface collects completed VLA proposals during a 500 ms eligibility window, freezes the candidate set and requests a selection from a multimodal Decisions API. The request includes synchronized camera views, labelled tool-path previews and timestamped simulator metadata. A timely valid choice is rechecked for state freshness and motion guards before execution from the selected plan's correct prefix. Unknown, stale, unsafe or timed-out replies cause an explicit hold/reobserve. Physics continues while waiting, with gripper commands preserved. Requests occur at planning boundaries.

An earlier asynchronous prototype receives 56 valid replies and activates zero selections. Of those replies, 55 abstain and 48 miss the observation-to-choice deadline. Three fallback chunks execute and the task fails. The revised implementation removes silent first-admissible movement. In a bounded real-model check, its 500 ms window admits one complete plan. Proposal generation takes 479–483 ms, including 178–181 ms of model inference. A mocked abstention produces holds only. One live reply selects the plan after 2.04 s and 27 selected controls execute before the test cap. The bounded test records API latency, plan activation and executed controls.

Offline future-view ranking is a separate privileged diagnostic. Across 12 decisions, a translucent-arm display yields two adequate choices among three selections and nine abstentions. Rendered future RGB yields five among five selections and seven abstentions. The rendered futures are produced by the simulator. Most requests abstain, leaving few selected actions for comparison.

5 Discussion

The experiments connect distinct teaching choices to specific behaviours. Source improves local grasping and placement. Range expands destination coverage. Counterbalanced task teaching removes a fixture-visibility shortcut. Acquisition remains a separate bottleneck, with fingertip contact and wrist height influencing whether the object stays upright through closure and lift.

The recordings give the study a task-specific foundation: observable starts and goals, deliberate orientation corrections, interrupted movements and examples where work is already complete. Their reconstructed scenes support robot-supervised adaptation. The visual-retargeting route additionally uses object motion from the footage, with endpoint registration and supplied acquisition/release control. Ordered, shuffled, static and endpoint-only variants all succeed on the inspected correction, leaving the value of intermediate motion unresolved.

The continuous evaluation covers two related scenes, fixed cameras/materials and frozen checkpoints. Its intervals quantify local scenario variation. The source-family comparisons likewise reuse six takes. One upstream mismatch reduces exact loading-only comparisons to 35 pairs. Uncalibrated physical scale and sampled contact logging remain practical limits.

Recovery and reversal need further teaching. Two learned reverse-task attempts fail, while two expert endpoint completions violate contact qualification. Small recovery-data additions leave the quarantined states unsolved. The next tests should prioritize calibrated collection sessions, qualified acquisition recoveries, broader holdouts and independently repeated training.

6 Reproducibility and availability

The source package contains the manuscript, plotting code, derived outcome tables, figure inputs and checksums. Figures regenerate from saved measurements. The simulation uses MuJoCo 2.3.7 and robosuite 1.4.1, a 2 ms Euler physics step, Newton solver and 20 Hz control. Final stability scoring uses 12 samples at 10 Hz over 1.1 s.

Expanded temporal-model training used four NVIDIA H100 80 GB GPUs. Evaluation ran on an RTX 4090. Policy comparisons pause simulated time during inference. The online ranking prototype advances physics during waits and logs latency and state drift separately.

Independent saved-record rescoring reproduces every strict stage decision and workflow qualification in the final panel. The included package reproduces the analysis and presentation. Raw recordings, policy weights and the executable research repository are retained separately.

7 Conclusion

Personal recordings of feeding, reorientation, pauses and completed work supply reproducible scenarios for robot teaching and evaluation. Source adaptation establishes local manipulation, goal diversity improves destination coverage, and loading-specific teaching enables a continuous two-stage workflow. Controller assistance raises observed qualified success from 69.4% to 86.1% in the paired simulator panel. The remaining failures concentrate on acquisition, contact and joint-limit violations, and unfamiliar starts.

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